

Math 1232 Summer 2025

Mastery Quiz 7

Due Thursday, July 24

This mastery quiz has two topics. Everyone should do S6, but if you already have a 2/2 on S5, feel free to skip it. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write “yes” or “no” or give a single number.

Please turn in this quiz in class on Thursday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Secondary Topic 5: Differential Equations
- Secondary Topic 6: Sequences and Series

Name: Solutions

Secondary Topic 5: Differential Equations

(a) Find a general solution to the equation $y' = x^2 + 1 + x^2y + y$.

Solution. We have $y' = \frac{x^3}{2} + \frac{-1}{2x^3}$, and thus

$$\begin{aligned}\frac{dy}{dx} &= (x^2 + 1)(1 + y) \\ \int \frac{dy}{y} &= \int x^2 + 1 \, dx \\ \ln |1 + y| &= \frac{x^3}{3} + x + C \\ 1 + y &= e^{x^3/3+x+C} \\ y &= e^{x^3/3+x+C} - 1 = Ce^{x^3/3+x} - 1\end{aligned}$$

(Note in the end our last constant C is not the same as the one we started with.)

(b) Find the (particular) solution to $y' = \frac{3y^2}{x}$ if $y(1) = 1$.

Solution.

$$\begin{aligned}\frac{dy}{dx} &= \frac{3y^2}{x} \\ \int \frac{dy}{3y^2} &= \int \frac{1}{x} \, dx \\ \frac{-1}{3y} &= \ln |x| + C \\ 3y &= \frac{-1}{\ln |x| + C} \\ y &= \frac{-1}{3 \ln |x| + C}.\end{aligned}$$

Plugging in $x = 1, y = 1$ gives

$$\begin{aligned}1 &= \frac{-1}{3 \cdot 0 + C} = \frac{-1}{C} \\ C &= -1 \\ y &= \frac{-1}{3 \ln |x| - 1}.\end{aligned}$$

Secondary Topic 6: Sequences and Series

(a) Let $b_n = \frac{(n)!}{(n+2)!}$. Compute the first four terms of the sequence, and compute $\lim_{n \rightarrow \infty} b_n$.

Solution. $b_1 = \frac{1}{6}, b_2 = \frac{2}{24} = \frac{1}{12}, b_3 = \frac{6}{120} = \frac{1}{20}$, and $b_4 = \frac{24}{720} = \frac{1}{30}$.

We compute

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{(n+1)(n+2)} = \lim_{n \rightarrow \infty} \frac{1}{n^2 + 3n + 2} = 0.$$

(b) Compute $\sum_{n=1}^{\infty} \frac{2}{n^2 + 5n + 6}$. Remember to justify your computation rigorously.

Solution. A partial fractions decomposition tells us that $\frac{2}{n^2 + 5n + 6} = \frac{2}{n+2} - \frac{2}{n+3}$. Then our partial sums are

$$\begin{aligned} s_k &= \sum_{n=1}^k \frac{2}{n^2 + 5n + 6} \\ &= \left(\frac{2}{3} - \frac{2}{4} \right) + \left(\frac{2}{4} - \frac{2}{5} \right) + \cdots + \left(\frac{2}{k+2} - \frac{2}{k+3} \right) = \frac{2}{3} - \frac{2}{k+3} \\ \sum_{n=1}^{\infty} \frac{2}{n^2 + 5n + 6} &= \lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} \frac{2}{n^2 + 5n + 6} \\ &= \lim_{k \rightarrow \infty} \frac{2}{3} - \frac{2}{k+3} = \frac{2}{3}. \end{aligned}$$

(c) Compute $\sum_{n=1}^{\infty} \frac{4}{3^{2n}}$.

Solution. This is a geometric series with $a = 4/9$ and $r = 1/9$, so we have

$$\sum_{n=1}^{\infty} \frac{4}{3^{2n}} = \frac{4/9}{1 - 1/9} = 1/2.$$